Van-Giang Trinh: Curriculum Vitae

1 Personal



- Name: Van-Giang Trinh (Trịnh Văn Giang in Vietnamese)
- Address: No. 18, Hoang Quoc Viet Street, Nghia Do Ward, Hanoi City, Vietnam
- Email: tvgiang@math.ac.vn
- Google Scholar, Website
- Citizenship: Vietnamese

2 Employment History and Education

2.1 Employment History

- Researcher, Institute of Mathematics, Vietnam Academy of Science and Technology (VAST), Hanoi, Vietnam 11/2026–
- Lecturer, Faculty of Computer Science and Engineering, Ho Chi Minh City University of Technology 09/2025–10/2026
- Postdoc, Lifeware team, Inria Saclay, Palaiseau, France 10/2024–08/2025

  Supervisor: Research Director (DRCE). François Fages
- Postdoc, Laboratoire d'Informatique & Systèmes, Aix-Marseille University 04/2022–09/2024

  Supervisor: Prof. Belaid Benhamou
- Part-time Researcher, School of Information Science, Japan Advanced Institute of Science and Technology 01/2022–03/2022
- Doctoral Student, School of Information Science, Japan Advanced Institute of Science and Technology 10/2018–12/2021
- Research Student, School of Information Science, Japan Advanced Institute of Science and Technology 10/2017–09/2018
- Research Assistant, Faculty of Computer Science and Engineering, Ho Chi Minh City University of Technology 06/2016–06/2017
- Internship, School of Information Science, Japan Advanced Institute of Science and Technology 10/2015–11/2015

2.2 Education

- **Ph.D.**, Information Science, Japan Advanced Institute of Science and Technology, Japan 10/2018–12/2021

  Supervisor: Prof. Kunihiko Hiraishi

  Thesis Title: On attractor detection and optimal control of Boolean networks

  Jury: Prof. Kunihiko Hiraishi, Prof. Tatsuya Akutsu, Assoc. Prof. Koichi Kobayashi, Prof. Mineo Kaneko, and Assoc. Prof. Daisuke Ishii
- **Master of Science**, Computer Science, Ho Chi Minh City University of Technology, Vietnam 08/2014–04/2017

  Supervisor: Assoc. Prof. Quan Thanh Tho

- **Bachelor of Science**, Computer Science, Ho Chi Minh City University of Technology, Vietnam 09/2009–04/2014

  Supervisor: Assoc. Prof. Quan Thanh Tho

- **High School**, Le Loi High School, Tho Xuan, Thanh Hoa, Vietnam 09/2006–05/2009

3 Research

3.1 Research Interests

My research lies at the intersection of theoretical computer science, mathematical logic, symbolic artificial intelligence, and discrete dynamical systems. I investigate the semantics, structural properties, and computational complexity of Boolean networks, logic programming, formal argumentation, and Petri nets. A central objective is to uncover and exploit connections among these formalisms to develop new theoretical foundations and symbolic computational methods. My interests also extend to computational systems biology and constraint-based optimization.

3.2 Awards and Fellowships

1. CIMPA–ICRTM Fellowship Laureate, Thematic Semester "Graphs and Beyond", Vietnam 08/2026
2. Diversity and Inclusion (D&I) Grant, 23rd International Conference on Principles of Knowledge Representation and Reasoning (KR 2026) 05/2026
3. Outstanding Performance Award, Japan Advanced Institute of Science and Technology 12/2021
4. Japanese Government (Monbukagakusho: MEXT) Scholarship, Japan Advanced Institute of Science and Technology 10/2017–09/2021
5. JAIST President Award, Japan Advanced Institute of Science and Technology 09/2019

3.3 Publications

3.3.1 International Journals (top venues in bold)

1. Samuel Pastva, Kyu Hyong Park, Jordan C. Rozum, **Van-Giang Trinh**, & Réka Albert. (2026, August). Relating biomarkers and phenotypes using dynamical trap spaces. ***npj Systems Biology and Applications***. (Scimago Q1, IF 4.4) https://doi.org/10.1038/s41540-026-00807-y

2. **Van-Giang Trinh**, Kyu Hyong Park, Samuel Pastva, & Jordan C Rozum. (2025). Mapping the attractor landscape of Boolean networks with biobalm. ***Bioinformatics***, *41*(5). (Scimago Q1, IF 5.4, Google Scholar top 1/20) https://doi.org/10.1093/bioinformatics/btaf280

3. **Van-Giang Trinh**, Belaid Benhamou, & Sylvain Soliman. (2023b, September). Trap spaces of Boolean networks are conflict-free siphons of their Petri net encoding. ***Theoretical Computer Science***, *971*, 114073. (Scimago Q2, IF 1.1, CORE Rank A, Google Scholar top 5/20) https://doi.org/10.1016/j.tcs.2023.114073

4. Tarek Khaled, Belaid Benhamou, & **Van-Giang Trinh**. (2023, July). Using answer set programming to deal with Boolean networks and attractor computation: Application to gene regulatory networks of cells. ***Annals of Mathematics and Artificial Intelligence***, 1–38. (Scimago Q3, IF 1.2, CORE Rank C) https://doi.org/10.1007/s10472-023-09886-7

5. **Van-Giang Trinh**, Belaid Benhamou, Thomas Henzinger, & Samuel Pastva. (2023a, June). Trap spaces of multi-valued networks: Definition, computation, and applications. ***Bioinformatics***, *39*(Supplement_1), i513–i522. (Scimago Q1, IF 5.8, Google Scholar top 1/20) https://doi.org/10.1093/bioinformatics/btad262

6. **Van-Giang Trinh**, & Kunihiko Hiraishi. (2020b). On attractor detection and optimal control of deterministic generalized asynchronous random Boolean networks. ***IEEE/ACM Transactions on Computational Biology and Bioinformatics***, *19*(3), 1794–1806. (Scimago Q2, IF 4.5, CORE Rank C, Google Scholar top 7/20) https://doi.org/10.1109/TCBB.2020.3043785

7. **Van-Giang Trinh**, Tatsuya Akutsu, & Kunihiko Hiraishi. (2020). An FVS-based approach to attractor detection in asynchronous random Boolean networks. ***IEEE/ACM Transactions on Computational Biology and Bioinformatics***, *19*(2), 806–818. (Scimago Q2, IF 4.5, CORE Rank C, Google Scholar top 7/20) https://doi.org/10.1109/TCBB.2020.3028862

8. **Van-Giang Trinh**, & Kunihiko Hiraishi. (2020c). A study on attractors of generalized asynchronous random Boolean networks. *IEICE TRANSACTIONS on Fundamentals of Electronics, Communications and Computer Sciences*, *103*(8), 987–994. (Scimago Q3, IF 0.3, CORE Rank C) https://doi.org/10.1587/transfun.2019EAP1163

9. **Van-Giang Trinh**, Le Ngoc Kim Khanh, Bang Ngoc Bao Tam, Tram Loi Quan, Bui Hoai Thang, & Quan Thanh Tho. (2016). Modelling and congestion detection of wireless sensor networks: A concurrent-based approach using coloured Petri nets. *International Journal of Applied Information Systems*, *11*(7), 1–9. https://doi.org/10.5120/ijais2016451629

10. Pham Hong Long, **Van-Giang Trinh**, Dinh Hoang Mai, Mai Phuong Nam, Quan Thanh Tho, & Ngo Quang Hung. (2014). Assisting students in finding bugs and their locations in programming solutions. *International Journal of Quality Assurance in Engineering and Technology Education (IJQAETE)*, *3*(2), 12–27. https://doi.org/10.4018/ijqaete.2014040102

3.3.2 International Conferences (top venues in bold)

1. Ngoc-Mai Pham, Trang-Linh Nguyen, Thi-Hai-Yen Vuong, Ha-Thanh Nguyen, & **Van-Giang Trinh**. (2026, July). Walk-in multi-stage patient flow scheduling: An ASP model with DES-based evaluation. In ***International Conference on Logic Programming***. (regular paper, top-one in logic programming) https://doi.org/10.4204/EPTCS.450.23

2. Samuel Pastva, & **Van-Giang Trinh**. (2026). BAss: Symbolic reasoning in abstract dialectical frameworks. In ***International Conference on Principles of Knowledge Representation and Reasoning***. (accepted, regular paper, AR 25%, CORE Rank A*, top-one in knowledge representation and reasoning) https://doi.org/10.24963/kr.2026/70

3. **Van-Giang Trinh**, Sylvain Soliman, François Fages, & Belaid Benhamou. (2025, September). On the trap space semantics of normal logic programs. In ***International Conference on Logic Programming***. (regular paper, CORE Rank A, top-one in logic programming) https://doi.org/10.4204/EPTCS.439.21

4. Mohimenul Kabir*, **Van-Giang Trinh***, Samuel Pastva*, & Kuldeep S. Meel. (2025, August). Scalable counting of minimal trap spaces and fixed points in Boolean networks. In ***International Conference on Principles and Practice of Constraint Programming***. (regular paper, * = equal contribution, AR 39%, CORE Rank A, top-one in constraint programming) https://doi.org/10.4230/LIPIcs.CP.2025.17

5. **Van-Giang Trinh**, Samuel Pastva, Jordan C Rozum, Kyu Hyong Park, & Réka Albert. (2025, June). On the number of asynchronous attractors in AND-NOT Boolean networks. In ***International Workshop on Cellular Automata and Discrete Complex Systems***. (exploratory paper) https://doi.org/10.48550/arXiv.2503.19147

6. **Van-Giang Trinh**, Belaid Benhamou, & Vincent Risch. (2025, February). Graphical analysis of abstract argumentation frameworks via Boolean networks. In *International Conference on Agents and Artificial Intelligence* (pp. 745–756). (regular paper, AR 23.09%, CORE Rank B) https://doi.org/10.5220/0013346400003890

7. Quang-Anh Nguyen, Thu-Trang Pham, Thi-Hai-Yen Vuong, **Van-Giang Trinh**, & Ha-Thanh Nguyen. (2025, February). Detecting misleading information with LLMs and explainable ASP. In *International Conference on Agents and Artificial Intelligence* (pp. 1327–1334). (short paper, CORE Rank B) https://doi.org/10.5220/0013357400003890

8. **Van-Giang Trinh**, Belaid Benhamou, Sylvain Soliman, & François Fages. (2024, October). Graphical conditions for the existence, unicity and number of regular models. In ***International Conference on Logic Programming*** (pp. 175–187). (regular paper, CORE Rank A, top-one in logic programming) https://doi.org/10.4204/EPTCS.416.16

9. **Van-Giang Trinh**, Belaid Benhamou, Samuel Pastva, & Sylvain Soliman. (2024, February). Scalable enumeration of trap spaces in Boolean networks via answer set programming. In ***Annual AAAI Conference on Artificial Intelligence*** (p. 10714-10722). (regular paper, **oral presentation**, AR 23.8% for the overall conference and 2.3% for oral presentation, CORE Rank A*, top-one in general AI) https://doi.org/10.1609/aaai.v38i9.28943

10. **Van-Giang Trinh**, Belaid Benhamou, & Sylvain Soliman. (2023a, August). Efficient enumeration of fixed points in complex Boolean networks using answer set programming. In ***International Conference on Principles and Practice of Constraint Programming*** (pp. 35:1–35:19). (regular paper, AR 40%, CORE Rank A, top-one in constraint programming) https://doi.org/10.4230/LIPIcs.CP.2023.35

11. **Van-Giang Trinh**, Belaid Benhamou, Thomas Henzinger, & Samuel Pastva. (2023b, July). Trap spaces of multi-valued networks: Definition, computation, and applications. In ***The 31st Annual Intelligent Systems For Molecular Biology and the 22nd Annual European Conference on Computational Biology***. Oxford University Press. (regular paper, AR 17.9%, CORE Rank A, top-one in computational biology) https://doi.org/10.1093/bioinformatics/btad262

12. **Van-Giang Trinh**, Kunihiko Hiraishi, & Belaid Benhamou. (2022, August). Computing attractors of large-scale asynchronous Boolean networks using minimal trap spaces. In ***ACM International Conference on Bioinformatics, Computational Biology and Health Informatics*** (pp. 1–10). ACM. (regular paper, AR 29%, flagship conference of the ACM SIGBio) https://doi.org/10.1145/3535508.3545520

13. **Van-Giang Trinh**, Belaid Benhamou, Kunihiko Hiraishi, & Sylvain Soliman. (2022, August). Minimal trap spaces of logical models are maximal siphons of their Petri net encoding. In *International Conference on Computational Methods in Systems Biology* (pp. 158–176). Springer. (regular paper, AR 65%) https://doi.org/10.1007/978-3-031-15034-0_8

14. **Van-Giang Trinh**, & Kunihiko Hiraishi. (2021, October). An improved method for finding attractors of large-scale asynchronous Boolean networks. In *IEEE International Conference on Computational Intelligence in Bioinformatics and Computational Biology* (pp. 1–9). IEEE. (regular paper, AR 51%, CORE Rank C) https://doi.org/10.1109/cibcb49929.2021.9562947

15. **Van-Giang Trinh**, & Kunihiko Hiraishi. (2019, June). Algorithms for finding attractors of generalized asynchronous random Boolean networks. In *12th Asian Control Conference* (pp. 67–72). IEEE. Retrieved from `http://ieeexplore.ieee.org/document/8765169` (regular paper, AR 73%)

16. Le Ngoc Kim Khanh, **Van-Giang Trinh**, Bui Hoai Thang, & Quan Thanh Tho. (2017, April). Probabilistic modelling for congestion detection on wireless sensor networks. In *International Conference on Control, Decision and Information Technologies* (pp. 0190–0195). IEEE. (regular paper, AR 48%, CORE Rank C) https://doi.org/10.1109/CoDIT.2017.8102589

17. **Van-Giang Trinh**, Nguyen Duc Khoan, Nguyen Duy Khuong, Vu Phu Thuc, & Quan Thanh Tho. (2016, September). Fast-and-Fit: An intelligent auto-pricing system for airlines travel agencies. In *SAI Intelligent Systems Conference* (pp. 853–865). Springer. (regular paper) https://doi.org/10.1007/978-3-319-56994-9_58

18. **Van-Giang Trinh**, Kunihiko Hiraishi, & Quan Thanh Tho. (2016, July). Modeling and analysing Boolean networks by coloured Petri nets. *IEICE Proceedings Series*, *61*(4447). (regular paper) https://doi.org/10.34385/proc.61.4447

19. Bao Trung Pham Duy, **Van-Giang Trinh**, Le Dinh Thuan, & Quan Thanh Tho. (2015, October). Reusing symbolic observation graph for efficient model checking. In *International Conference on Knowledge and Systems Engineering* (pp. 250–255). IEEE. (regular paper) https://doi.org/10.1109/kse.2015.44

3.3.3 International Workshops

1. **Van-Giang Trinh**, & Kunihiko Hiraishi. (2020a, December). An efficient method for approximating attractors in large-scale asynchronous Boolean models. In *International Workshop on Biological Network Analysis and Integrative Graph-Based Approaches, in conjunction with IEEE International Conference on Bioinformatics and Biomedicine* (pp. 1820–1826). IEEE. (workshop paper, AR 60%) https://doi.org/10.1109/bibm49941.2020.9313230

3.3.4 Technical Reports

1. **Van-Giang Trinh**, & Belaid Benhamou. (2024). Static analysis of logic programs via Boolean networks. *CoRR*, *abs/2407.09015*. https://doi.org/10.48550/ARXIV.2407.09015

2. **Van-Giang Trinh**, Belaid Benhamou, & Loïc Paulevé. (2024). mpbn: a simple tool for efficient edition and analysis of elementary properties of Boolean networks. *CoRR*, *abs/2403.06255*. https://doi.org/10.48550/ARXIV.2403.06255

3.3.5 Other Publications

1. F. Kordon, H. Garavel, L. M. Hillah, F. Hulin-Hubard and G. Chiardo, A. Hamez, L. Jezequel, A. Miner, J. Meijer, E. Paviot-Adet, D. Racordon, C. Rodriguez, C. Rohr, J. Srba, Y. Thierry-Mieg, **Van-Giang Trinh**, & K. Wolf. (2016, June). *Complete Results for the 2016 Edition of the Model Checking Contest*. Retrieved from `https://mcc.lip6.fr/2016/index.php`

3.4 Talks and Seminars

1. Thematic Semester "Graphs and Beyond" (Vietnam Institute of Mathematics, Hanoi, Vietnam), From Graph Structure to Argumentation Semantics: A Dynamics-Based Analysis of Abstract Argumentation Frameworks (11/08/2026)

2. PTIT seminar (Hanoi, Vietnam), Scalable enumeration of trap spaces in Boolean networks via answer set programming (28/08/2025)

3. HUS seminar (Hanoi, Vietnam), Scalable enumeration of trap spaces in Boolean networks via answer set programming (27/08/2025)

4. Joint seminar of the Department of Discrete Mathematics, Probabilily Theory and Statistics, and the ICRTM (Hanoi, Vietnam), Connecting Abstract Argumentation Frameworks and Boolean Networks (12/08/2025)

5. MLLab seminar (Hanoi, Vietnam), Scalable enumeration of trap spaces in Boolean networks via answer set programming (07/08/2025)

6. LIRICA Seminar (St Charles, Marseille), On the Trap Space Semantics of Normal Logic Programs (26/06/2025)

7. LIRICA Seminar (St Charles, Marseille), Graphical conditions for the existence, unicity and number of regular models (25/11/2024)

8. MABioS seminar (Luminy, Marseille), Trap spaces of multi-valued networks: definition, computation, and applications (30/09/2024)
9. Seminar on Mathematical Foundations for Computer Science (Hanoi, Vietnam), Trap spaces of Boolean networks are conflict-free siphons of their Petri net encoding (20/08/2024)
10. UET seminar (Hanoi, Vietnam), Scalable enumeration of trap spaces in Boolean networks via answer set programming (19/08/2024)
11. CSE-HCMUT seminar (Ho Chi Minh city, Vietnam), Scalable enumeration of trap spaces in Boolean networks via answer set programming (12/08/2024)
12. Journées annuelles 2024 du GT Bioss (LIP6, Paris), Trap spaces of multi-valued networks: definition, computation, and applications (27/05/2024)
13. LIRICA Seminar (St Charles, Marseille), Scalable enumeration of trap spaces in Boolean networks via answer set programming (25/03/2024)
14. Lifeware Seminar (INRIA Saclay), Trap spaces of Boolean networks are conflict-free siphons of their Petri net encoding (05/12/2023)
15. IRISA Seminar (IRISA, Rennes), Efficient enumeration of fixed points in complex Boolean networks using answer set programming (30/11/2023)
16. LIRICA Seminar (St Charles, Marseille), Efficient enumeration of fixed points in complex Boolean networks using answer set programming (20/11/2023)
17. Demi Journées du Pôle Calcul on Artificial Intelligence (Luminy, Marseille), Efficient enumeration of fixed points in complex Boolean networks using answer set programming (15/06/2023)
18. CANA Seminar (Luminy, Marseille), Trap spaces of Boolean networks are conflict-free siphons of their Petri net encoding (30/05/2023)
19. Journées BioLogique BIOSS/CAVIAR (LIP6, Paris), An approach based on ASP and Petri nets for the calculation of attractors in Boolean networks (25/05/2023)
20. Journées Scientifiques du LIS (Carry le Rouet), Efficient enumeration of minimal trap spaces in large-scale Boolean networks of gene networks (23/05/2023)
21. LIRICA Seminar (St Charles, Marseille), Minimal trap spaces of Boolean models are maximal siphons of their Petri net encoding (17/10/2022)
22. LIRICA Seminar (Virtual), An FVS-based approach to attractor detection in asynchronous Boolean networks (28/06/2021)

4 Teaching

4.1 Courses

- Discrete Structure of Computing (undergraduate course, lecturer)
  - Ho Chi Minh City University of Technology: 01/2026–05/2026
- Principles of Programming Languages (undergraduate course, lecturer)
  - Ho Chi Minh City University of Technology: 01/2026-05/2026
- Mathematical Modeling (undergraduate course, lecturer)
  - Ho Chi Minh City University of Technology: 08/2025–12/2025
- Exercises on Graph Theory (graduate course, teaching assistant)
  - Japan Advanced Institute of Science and Technology: 06/2021–08/2021

- Functional Programming (graduate course, teaching assistant)
  - Japan Advanced Institute of Science and Technology: 10/2020–12/2020
- Introduction to Computer Programming (undergraduate course, visiting lecturer)
  - Ho Chi Minh City University of Technology: 01/2017–06/2017
- Principles of Programming Languages (undergraduate course, teaching assistant)
  - Ho Chi Minh City University of Technology: 09/2013–01/2014, 09/2014–01/2015, 09/2015–01/2016
- Object-Oriented Programming (undergraduate course, teaching assistant)
  - Ho Chi Minh City University of Technology: 09/2015–01/2016
- Principles of Programming Languages (undergraduate course, visiting lecturer)
  - Ho Chi Minh City University of Natural Resources and Environment: 01/2015–05/2015
- Data Structures and Algorithms (undergraduate course, teaching assistant)
  - Ho Chi Minh City University of Technology: 09/2012–01/2013, 01/2015–05/2015
- Programming Fundamentals (undergraduate course, teaching assistant)
  - Ho Chi Minh City University of Technology: 01/2015–05/2015

5 Professional Services

5.1 Program Committee Memberships

- International Conference on Logic Programming and Non-monotonic Reasoning 2026
- International Conference on Principles of Knowledge Representation and Reasoning 2026
- International Workshop on Nonmonotonic Reasoning 2026
- International Workshop on Nonmonotonic Reasoning 2025
- Workshop on Answer Set Programming and Other Computing Paradigms 2026
- Workshop on Answer Set Programming and Other Computing Paradigms 2025

5.2 Journal Reviewers

- Bioinformatics (2026—1 paper)
- Nonlinear Dynamics (2026—1 paper)
- SIAM Journal on Applied Dynamical Systems (2023—1 paper)
- IEEE/ACM Transactions on Computational Biology and Bioinformatics (2022—1 paper, 2023—1 paper, 2026—1 paper)

5.3 Conference Reviewers

- International Conference on Logic Programming 2025 (subreviewer—1 paper)
- International Conference on Computational Methods in Systems Biology 2025 (subreviewer—1 paper)

6 References

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